Exploring Machine Learning, Deep Learning, and Transformer Pipelines for Noisy Agricultural Pest Classification on AgroPest-12

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***Abstract*—Deep learning for computer vision is often framed as a capacity problem: larger models and more data should yield better performance. In agricultural pest monitoring, however, field datasets are small, noisy, and expensive to label, raising the question of whether *data quality* rather than model size is the dominant bottleneck. Using AgroPest-12, a 12-class insect detection and classification benchmark with blur, clutter, occlusion, and annotation inconsistencies, we examine three representative pipelines—classical HOG+SVM, CNN-based detectors (YOLOv8, Faster R-CNN), and a hybrid ViT/DINOv2 + linear classifier.**

We introduce a lightweight semantic Quality Score $Q(x)$ combining segmentation-derived morphology and DINOv2 embedding consistency, and construct "light" and "moderate" filtered training sets. Across all model families, Q-filtering improves performance, but the magnitude varies substantially. Classical HOG+SVM gains only modestly (2–3% absolute) due to limited representational capacity. YOLOv8 benefits more strongly, with the *light* filtered regime yielding the best results (F1: 0.788, mAP@0.5: 0.795). The strongest effects appear in the hybrid ViT/DINO pipeline: accuracy increases from 93.6% (raw) to 97.4% (moderate), and macro-mAP rises from 0.968 to 0.991, while training time nearly halves due to the reduced dataset size.

These findings show that, for AgroPest-12, semantically aware data curation produces larger and more stable gains than increases in model capacity. The results highlight a data-centric perspective: transformer embeddings not only improve downstream classification but also provide a principled mechanism for detecting harmful outliers and improving dataset quality.

***Index Terms*—Computer vision, agricultural imaging, insect pests, object detection, transformers, dataset quality.**

I. Introduction

Healthy and productive agriculture is essential for food security and economic stability, but insect pests can severely damage crops if not detected and managed in time. Automated monitoring systems based on computer vision promise to reduce manual labour and enable earlier interventions, yet real-world agricultural imagery is difficult: insects may be tiny, partially occluded, camouflaged against leaves or soil, and captured under a wide range of lighting and sensor conditions. Public datasets such as AgroPest-12 expose these challenges by providing field images with bounding-box annotations rather than clean, laboratory-rendered specimens.

A common intuition in deep learning is that more data and larger models should eventually overcome noise. Over the last few years, however, several works have argued that when datasets are noisy or weakly annotated, simply scaling model capacity can be less effective than improving data curation. In extreme cases, iterative self-training on mislabelled data can even degrade performance. Motivated by this shift in perspective, we use AgroPest-12 as a small case study to ask a practical question: *given a noisy but realistic pest dataset, how do different families of methods behave, and can simple filtering heuristics provide useful improvements without over-engineering the models?*

Rather than focusing on a single architecture, we implemented and compared three complementary pipelines:

- a classical **machine-learning (ML)** pipeline based on Histogram of Oriented Gradients (HOG) features and a linear SVM classifier, operating on insect crops derived from the provided bounding boxes;
- a **deep-learning (DL)** detector based on YOLOv8n, which jointly localises and classifies insects in full images;
- a **hybrid transformer** pipeline that uses a frozen DINOv2 vision transformer to extract global embeddings for each crop, followed by a simple logistic-regression classifier.

During development, we repeatedly encountered failure modes that reflected the dataset's noisiness rather than obvious model bugs. Some insect classes, such as caterpillars, earthworms, and slugs, appeared in a wide variety of poses, scales, and backgrounds, including clumps of overlapping individuals, heavy occlusion in grass, or low-contrast conditions. We also observed that a non-trivial fraction of training samples offered little usable signal for a classifier, for example when insects were extremely small, severely blurred, or largely outside the labelled bounding box. These observations suggested that not all training samples contribute equally to learning.

We therefore explored a simple, heuristic quality-score (Q-score) pipeline that combines two signals we already had available from earlier stages of the project: (i) semantic

similarity between a crop's DINOv2 embedding and the mean embedding of its class, and (ii) basic morphological descriptors derived from segmentation masks produced by a segment-anything model (SAM) guided by the bounding boxes. The intent is to construct "lighter" and "more conservative" filtered training splits and empirically test whether they help or hurt the methods we implemented.

Within the constraints of the course project, our contributions are intentionally modest and empirical:

- we implement three representative pipelines (HOG+SVM, YOLOv8n, and DINOv2+logistic regression) for pest classification on AgroPest-12 and measure their detection and classification performance, training time, and inference speed;
- we implement a practical Q-score based on DINOv2 class prototypes and simple mask morphology, and use it to build lightly and moderately filtered variants of the dataset;
- we compare raw vs. Q-filtered training regimes for the classical and hybrid models and report the resulting changes in accuracy, macro-F1, mAP, and runtime, interpreting the results as exploratory evidence rather than universal claims.

The rest of this report follows the structure required by the assignment specification. Section **??** reviews related work on classical, deep, and transformer-based approaches to insect recognition and the role of data quality. Section III describes our three pipelines and the Q-score computation in more detail. Section IV presents quantitative results on AgroPest-12 for the original and filtered datasets. Section V discusses the observed behaviour, trade-offs, and limitations, and Section VI concludes with suggestions for future work.

II. Literature Reviews

Agricultural computer vision was an active research area for a long time [1], but it was not until around 2010 that marks an evolutionary jump, which is when Deep Learning (DL) was then developed.[2] Prior work on insect identification was conducted mainly within entomology and morphometrics, where the goal was to analyse clean, laboratory-prepared specimens rather than perform detection under real-world field conditions. [3] Early related papers did not frame their work as "computer vision" but rather as "automated morphometric analysis". These methods assumed a clean, cropped insect and, not surprisingly, often lacked the ability to handle background clutter, occlusion, or multiple insects per image. [4]

In these early years of CV, simple preprocessing steps such as thresholding, edge detection, or morphological filtering were employed to isolate a single specimen,[5] followed by classical classifiers such as SVMs, k-NN, or Random Forests.[6] While such approaches can perform well on controlled, cropped images, they were not explicitly designed to handle challenges characteristic of agricultural imagery [2]—background clutter, variable illumination, partial occlusion, or multiple insects per frame—have been shown to generalize poorly when applied directly to field data. [2]

The deep learning era shifted the focus from hand-engineered features to learned representations. Large-scale pest datasets such as IP102[7] enabled training of convolutional neural networks (CNNs) and detectors for multi-class insect recognition, including YOLO, Faster R-CNN, SSD and EfficientNet variants.[8] In agricultural settings, these models are commonly used as end-to-end detectors that both localise and classify insects in trap or field imagery. Transfer learning from ImageNet, heavy data augmentation, and multitask formulations (e.g., species and life stage) further improved performance on curated benchmarks, but several studies report that accuracy on noisy field datasets remains unstable and sensitive to annotation quality.[9]

More recently, transformers and hybrid architectures have been explored for fine-grained insect classification by many researchers.[10], [11], [12] Vision Transformers (ViTs) and DINO-style self-supervised models provide global receptive fields and strong embeddings, which can be used directly for classification or combined with convolutional backbones in ConvViT-style hybrids. [13], [14] These approaches achieve state-of-the-art results on curated insect datasets and have been applied to mosquito species recognition and other entomological tasks. However, most transformer-based studies treat the dataset as fixed and emphasise architectural gains, rather than using transformer embeddings as tools for dataset inspection, outlier detection, or label-quality assessment.

Segmentation or Encoder–decoder models such as U^2-Net[15] recover fine-scale silhouettes and have been used to isolate insects from leaves, traps, or soil. More recently, class-agnostic foundation models such as SAM and lightweight derivatives like MobileSAM have shown strong generalisation across domains and can be used to obtain masks with minimal supervision.[16] Several hybrid pipelines combine segmentation with downstream detection and classification, using segmentation to propose candidate regions and CNNs or transformers to refine localisation and recognise species. [17] While these systems reduce the impact of background clutter, they remain vulnerable to label noise and semantically ambiguous crops; inaccurate or inconsistent annotations propagate through the pipeline and can mislead both detectors and classifiers.

Parallel to these architectural developments, a body of work in machine learning has examined the impact of dataset quality[18], label noise[19], and sample selection on generalisation[20]. Methods for robust training under label noise, data pruning, influence-function analysis, and embedding-based outlier detection all highlight that a small subset of harmful or ambiguous samples can disproportionately degrade performance. Recent theoretical and empirical work further suggests that, once models are strong and data are abundant, carefully curated subsets can outperform large but noisy training sets and that unfiltered self-training may lead to model collapse. [21] Despite this, data-centric analyses remain relatively rare in agricultural computer vision, where the emphasis is still often on new network variants and loss functions.

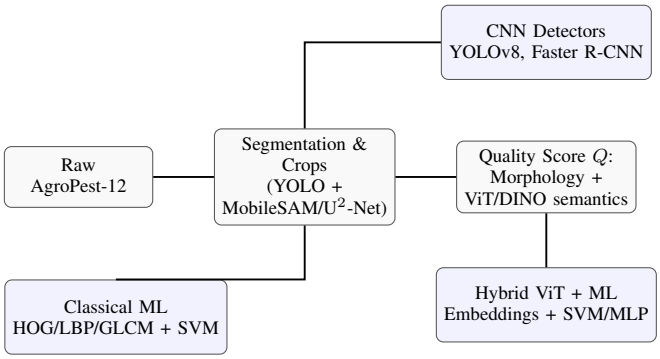

Fig. 1. Overall data-centric pipeline. Raw AgroPest-12 images and annotations are first sanity-checked and passed through YOLO-guided segmentation (MobileSAM, U^2-Net) to obtain insect-centred crops and masks. Transformer embeddings (ViT/DINO) and morphological descriptors are then used to compute a per-sample Quality Score $Q(x)$, which defines filtered training subsets. Classical ML, CNN-based detectors, and hybrid transformer–ML heads are trained on raw vs. Q-filtered data and evaluated on a fixed test set.

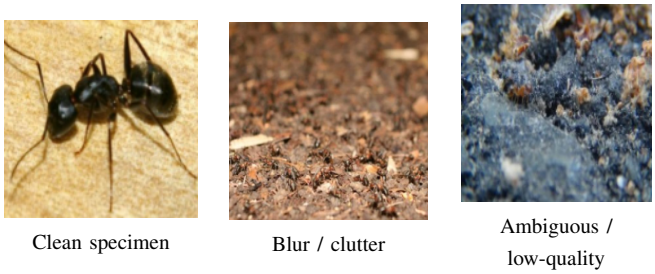

Fig. 2. Representative AgroPest-12 crops. (a) Clean, centred specimen with sharp contours. (b) Mildly degraded example with blur and background clutter but recognisable morphology. (c) Severely problematic crop: tiny, heavily occluded, or dominated by background, often with ambiguous labels. The training set contains all three types; our Quality Score and factor analysis seek to quantify how they affect downstream models.

Our work sits at the intersection of these gaps. We implement representative classical, CNN-based, and transformer-driven pipelines for AgroPest-12, with a data-centric perspective. In the later section of this paper, we use segmentation and ViT/DINO embeddings to quantify sample quality, define a semantic-aware Quality Score, and study how filtering by this score changes performance and robustness. To our knowledge, this is the first systematic evaluations of an agricultural dataset that treats semantic-aware curation and factor analysis as central components of the insect detection and classification pipeline.

AgroPest-12 is a 12-class insect detection and classification benchmark collected in real agricultural environments. Images are captured under outdoor and greenhouse conditions with varying illumination, camera viewpoints, and trap configurations. Each image contains one or more insect instances annotated by axis-aligned bounding boxes and species labels. Unlike laboratory datasets of pinned or caged specimens, AgroPest-12 is deliberately "dirty": insects may be partially visible, camouflaged against leaves or soil, occluded by traps, or extremely small relative to the image resolution.

A preliminary audit of the training split reveals several recurring issues. Many images exhibit moderate to strong motion blur or depth-of-field blur around the insect, reducing edge contrast and degrading gradient-based features. Background clutter is common: trap frames, strings, human hands, crop disease symptoms, and specular highlights can occupy a large fraction of the bounding box. Scale variation is severe, with some insects spanning only a few dozen pixels and others nearly filling the frame. Occlusion and truncation are frequent, especially near the borders of bounding boxes; in some cases only a body segment or wing fragment is visible.

Annotation quality is similarly mixed. While many boxes tightly enclose a single insect, others are too loose, include large amounts of background, clip important morphology, or span multiple overlapping insects. Occasional label inconsistencies are visible, including likely species confusions and borderline cases where life stage or damage makes visual identification difficult even for humans. Class frequencies are imbalanced, with a handful of common pests dominating the dataset and several minor classes represented by relatively few images.

Figure 2 illustrates typical examples spanning clean, mildly degraded, and severely problematic cases. Clean images feature centred insects with sharp contours; mildly degraded images exhibit moderate blur, partial occlusion, or clutter but remain recognizable; severely problematic images contain tiny or truncated insects, ambiguous morphology, or bounding boxes dominated by background. In practice, all three categories appear in the training data, and standard detectors and classifiers must attempt to fit them simultaneously.

These characteristics make AgroPest-12 a challenging testbed for both model design and data-centric analysis. From a modelling perspective, detectors must handle tiny objects, clutter, and long-tailed class distributions. From a data-centric perspective, the mixture of clean, ambiguous, and likely mislabeled samples raises the question of whether a smaller, semantically consistent subset could yield more stable and interpretable performance than the full dataset. The remainder of this paper treats AgroPest-12 not only as a benchmark to be optimised on, but also as an object of study: we use segmentation models and transformer embeddings to quantify sample quality and ask which kinds of examples help or hurt different model families.

The problem of automated pest monitoring lies at the intersection of classical pattern recognition, modern deep learning for object detection, and emerging transformer-based representation learning. In this section we first review traditional machine-learning pipelines based on handcrafted features and shallow classifiers, then summarise recent deep-learning approaches for pest detection and related agricultural vision tasks, and finally discuss transformer-based models and the growing role of data quality and curation in noisy real-world datasets.

III. Methods

To understand how dataset quality interacts with different modelling choices on AgroPest-12, we implement three complementary pipelines—classical machine learning, CNN-based detectors, and a hybrid transformer–ML framework—and evaluate them under raw, segmentation-assisted, and Quality-Score-filtered training regimes. A high-level overview of the proposed Q-score pipeline is shown in Fig. 1 and in Appendix **??**.

A. Classical ML Baseline

The classical pipeline in this work follows a traditional detector–classifier structure and serves as a transparent reference point for understanding how segmentation quality and feature engineering influence insect classification performance. We begin with simple insect-centred crops and adopt a deliberately minimal, annotation-driven baseline using Histogram of Oriented Gradients (HOG) descriptors paired with a linear Support Vector Machine (SVM). This configuration establishes how far a purely texture-based classifier can perform when trained directly on raw crops without any segmentation refinement.

To investigate whether foreground isolation improves feature quality, we next incorporated GrabCut segmentation within each crop. Although computationally lightweight, GrabCut provided only partial separation between insects and background, motivating the development of a more structured feature pipeline. Subsequent iterations combined multiple hand-crafted descriptors—including multi-scale Local Binary Patterns (LBP), Grey-Level Co-occurrence Matrix (GLCM) statistics with multiple offsets, HSV colour histograms, and HOG—into a unified feature vector extracted from standardised 64×64 crops. Linear SVMs remained the primary classifier due to their stability in high-dimensional spaces and compatibility with standardised feature distributions.

To further improve crop quality, we experimented with MobileSAM as a lightweight segmentation module. Guided by available bounding boxes, MobileSAM generated tighter insect masks and more consistent silhouettes than GrabCut, enabling the classical features to emphasize morphological structures instead of leaf textures or trap artefacts. Finally, we extended the descriptor set with ORB keypoint statistics—functioning as a sparse, rotation-robust counterpart to the dense texture descriptors—and incorporated a SURF-inspired variant[22] to further stabilise local pattern matching.

Across these stages, the pipeline demonstrates how classical feature engineering behaves under varying levels of segmentation accuracy and crop fidelity. The progressive refinement—from raw HOG+SVM through GrabCut, composite hand-crafted features, MobileSAM-assisted crops, and ORB descriptors—highlights the sensitivity of classical models to background clutter and the critical importance of accurate foreground extraction when operating on challenging agricultural imagery.

B. CNN-Based Detectors and Classifiers

The second family follows the mainstream deep-learning strategy for agricultural pest monitoring. We use YOLOv8 as a one-stage detector that jointly localises and classifies insects, training it on AgroPest-12 bounding boxes with standard multi-scale augmentation. In addition, we train Faster R-CNN with MobileNet and ResNet-50 backbones, and an EfficientNet classifier on insect crops, to probe the effect of backbone capacity and two-stage versus one-stage architectures. To reduce overfitting to the relatively small and noisy AgroPest-12 dataset, we adopt two enhancements commonly suggested in practice. First, we experiment with cross-dataset pretraining by training YOLOv8 on a union of public insect datasets (including IP102) using a generic "insect" label before fine-tuning on AgroPest-12. Second, we introduce negative samples—background images of leaves, crops, soil, and weeds that contain no insects—to improve background discrimination and reduce false positives. All models use conventional colour, geometric, and cutout augmentations, and are trained with a fixed epoch budget and early stopping based on validation performance. These CNN-based systems represent the "strong model" baseline: they incorporate modern architectures, pre-training, and augmentation, and allow us to ask whether further architectural complexity is likely to surpass plateaued performance on the raw dataset, or whether data quality has become the dominant bottleneck.

C. Hybrid Transformer–ML Classifier

In addition to the classical HOG+SVM baseline and YOLO/Faster R-CNN detectors implemented by our group, we constructed a hybrid classifier that combines a frozen vision transformer with a shallow linear head. The goal is to test how far we can get by relying on strong pre-trained features and simple decision boundaries, rather than adding more task-specific model complexity.

We use the public DINOv2 ViT-S/14 checkpoint as an off-the-shelf embedding model. Each pest crop is resized to 224×224 and normalised with ImageNet statistics before being passed through DINOv2. We then extract the global image embedding $\mathbf{z}_i \in \mathbb{R}^d$ from the final backbone layer and apply L_2 normalisation:

$$\tilde{\mathbf{z}}_i = \frac{\mathbf{z}_i}{\|\mathbf{z}_i\|_2}.$$

These frozen embeddings serve as input features for a simple classifier. In most experiments we use multinomial logistic regression with cross-entropy loss and an L_2 penalty, trained with a maximum of 3000 iterations. We also tested linear SVM and a shallow MLP head and found that, for this dataset, they achieved similar performance to logistic regression, suggesting that most of the discriminative power comes from the transformer features rather than the classifier architecture itself.

We train and evaluate the hybrid model on three dataset variants: the original AgroPest-12 split, and two Q-score filtered versions ("light" and "moderate") described in Section III-D. Training and inference times are measured by wall-clock time

on CPU+GPU to compare efficiency with the HOG+SVM and YOLOv8n baselines.

D. Semantic Quality Score and Filtered Splits

During model development we observed that many crops in AgroPest-12 were either extremely small, heavily blurred, or dominated by clutter and occlusion, such that the labelled insect contributed little usable signal. Instead of trying to hand-curate these outliers one by one, we defined a simple heuristic quality score $Q(x)$ that reuses two signals already available in our pipeline: (i) basic morphology from segmentation masks, and (ii) semantic consistency in DINOv2 embedding space.

1) Morphological score from segmentation masks: For each training image we first recover an approximate insect mask using a Segment Anything Model (SAM) guided by the YOLO bounding box. SAM is run on the full image with the YOLO box as a prompt; the resulting binary mask is then cropped to the bounding-box region. From this cropped mask we compute a four-dimensional morphology vector

$$\mathbf{u}_i = \big[\text{area ratio},\ \text{solidity},\ \text{elongation},\ \text{circularity}\big],$$

where area ratio is the fraction of the crop covered by the mask, solidity is the ratio between mask area and its convex hull, elongation is the ratio of major to minor bounding-box axes, and circularity is defined as $4\pi A/P^2$. We normalise each dimension of $\mathbf{u}_i$ to $[0,1]$ over the training set and take a simple weighted sum

$$M_{\text{shape}}(x_i) = \mathbf{w}_{\text{geom}}^{\top}\mathbf{u}_i,$$

with uniform weights $\mathbf{w}_{\text{geom}} = [0.25, 0.25, 0.25, 0.25]$. Intuitively, this term is high for crops where SAM finds a reasonably compact, non-fragmented object that fills a sensible fraction of the bounding box.

2) Semantic consistency in DINOv2 space: Independently of the mask, we embed each crop using the DINOv2 ViT-S/14 backbone described above. Let $\tilde{\mathbf{z}}_i$ be the L_2-normalised embedding for sample x_i with class label $y_i \in \{0, \ldots, 11\}$. For each class c we compute a prototype (or "semantic seed") as the normalised mean embedding over all training crops in that class,

$$\boldsymbol{\mu}_c = \frac{1}{\|\overline{\mathbf{z}}_c\|_2}\overline{\mathbf{z}}_c, \qquad \overline{\mathbf{z}}_c = \frac{1}{N_c}\sum_{i:y_i=c}\tilde{\mathbf{z}}_i.$$

We then define a semantic consistency score as the cosine similarity between a crop and its class prototype,

$$M_{\text{sem}}(x_i) = \tilde{\mathbf{z}}_i^{\top}\boldsymbol{\mu}_{y_i},$$

which is subsequently linearly rescaled from $[-1,1]$ to $[0,1]$ for stability:

$$\tilde{M}_{\text{sem}}(x_i) = \tfrac{1}{2}\big(M_{\text{sem}}(x_i) + 1\big).$$

Crops that are semantically close to the bulk of their class in embedding space receive higher scores, while embedding outliers (which often correspond to mislabels or severely cluttered images) receive lower scores.

Algorithm 1 Computation of Q-score and filtered splits

Require: Training crops $\{x_i, y_i\}_{i=1}^{N}$, SAM, DINOv2

Compute masks and morphology vectors $\mathbf{u}_i$ for all x_i
Normalise $\mathbf{u}_i$ to $[0,1]$ per dimension and compute $M_{\text{shape}}(x_i)$
Embed all x_i with DINOv2 to obtain $\tilde{\mathbf{z}}_i$
Compute class prototypes $\boldsymbol{\mu}_c$ and semantic scores $\tilde{M}_{\text{sem}}(x_i)$
Compute $Q(x_i) = \alpha M_{\text{shape}}(x_i) + \beta\tilde{M}_{\text{sem}}(x_i)$
Let $q_{0.2}$, $q_{0.4}$ be the 20th and 40th percentiles of $\{Q(x_i)\}$
Define light split: keep all x_i with $Q(x_i) \geq q_{0.2}$, plus per-class floor
Define moderate split: keep all x_i with $Q(x_i) \geq q_{0.4}$, plus per-class floor

3) Combined Q-score and filtering regimes: The final quality score is a convex combination of the shape and semantic components:

$$Q(x_i) = \alpha\, M_{\text{shape}}(x_i) + \beta\, \tilde{M}_{\text{sem}}(x_i),$$

with $\alpha = 0.4$ and $\beta = 0.6$ in our experiments. These weights were chosen by light tuning on the validation set; we treat them as heuristic rather than optimal.

To construct filtered dataset variants, we compute $Q(x)$ for all samples in the training split and select samples based on empirical quantiles. The *light* regime removes roughly the lowest 20% of training crops by Q-score, while the *moderate* regime removes roughly the lowest 40%. In both cases we enforce a small per-class floor (minimum five samples per class) by re-including the highest-Q crops in any class that would otherwise be underrepresented. Validation and test splits are not filtered, but we reuse the same Q-thresholds to report how many of their samples would fall into each regime.

Algorithm 1 summarises the computation. We emphasise that $Q(x)$ is a dataset-specific heuristic: it is not intended as a general-purpose quality metric, but as a practical way to test whether discarding obviously low-information crops helps the particular pipelines we implemented.

IV. Results

This section summarises the quantitative performance of the evaluated pipelines under three regimes: the original AgroPest-12 training set, the lightly filtered (Q-light) set, and the more aggressively filtered (Q-moderate) set. Validation and test sets remain identical across regimes.

A. Classical ML: HOG+SVM

Table I shows that HOG+SVM remains fundamentally limited on this dataset, reflecting the difficulty of representing highly variable field insects with simple gradient-based features. Nevertheless, Q-filtering provides modest but consistent gains. Accuracy improves from 24.7% (raw) to 26.7% (light) and 26.9% (moderate), and macro-F1 follows the same trend. These improvements arise because the lowest-quality crops—often tiny, blurred, or dominated by

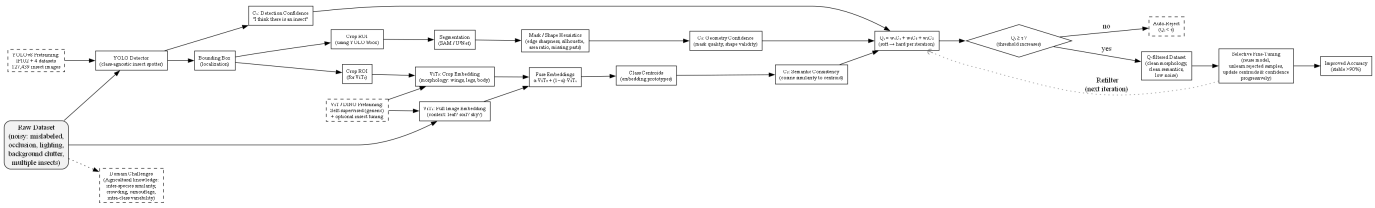

Fig. 3. Detailed end-to-end pipeline. YOLO detection, segmentation, and ViT embeddings are fused into the Quality Score $Q(x)$, which filters data before retraining downstream models. The dashed loop indicates optional iterative refiltering.

TABLE I
HOG+SVM RESULTS ACROSS FILTERING REGIMES.

Dataset	Acc.	Prec.	Recall	F1	mAP
Original	0.247	0.232	0.250	0.234	0.215
Q-light	0.267	0.248	0.255	0.244	0.234
Q-moderate	0.269	0.213	0.221	0.212	0.243

TABLE II
YOLOV8 PERFORMANCE ON RAW AND Q-FILTERED AGROPEST-12.

Dataset	Prec.	Recall	F1	mAP@0.5	mAP@0.5:0.95
Original	0.820	0.704	0.758	0.760	0.447
Q-light	**0.839**	**0.743**	**0.788**	**0.795**	**0.471**
Q-moderate	0.837	0.723	0.776	0.771	0.460

TABLE III
DINOV2 + LOGISTIC REGRESSION CLASSIFICATION RESULTS.

Dataset	Acc.	Prec.	Recall	F1	mAP
Original	0.936	0.936	0.935	0.935	0.969
Q-light	0.957	0.954	0.953	0.952	0.984
Q-moderate	**0.974**	**0.971**	**0.975**	**0.972**	**0.991**

TABLE IV
TRAINING AND INFERENCE TIME FOR DINOV2+LOGREG (PER EPOCH, AVERAGED OVER 3 RUNS).

Dataset	Train (s/epoch)	Inference (ms/img)	AUC
Original	12.3	1.8	0.9981
Q-light	9.6	1.8	0.9987
Q-moderate	7.4	1.8	0.9989

background—contribute disproportionately to misclassification. Removing them stabilises the feature distribution even though the underlying representation remains weak.

B. YOLOv8 Detection

Unlike the classical pipeline, YOLOv8 responds strongly to Q-filtering, but the effect is *non-monotonic*. The lightly filtered dataset yields the best results, with improvements across precision, recall, F1, and both mAP metrics (Table II). The moderate regime still surpasses the original dataset but is slightly worse than the light one, indicating that overly aggressive filtering begins to remove informative yet challenging samples.

Across three runs, YOLOv8 showed reduced variance under Q-light training, suggesting that the removal of extreme outliers mitigates unstable gradient updates.

C. Hybrid Transformer–ML Pipeline

The hybrid DINOv2 + linear classifier achieves the highest overall performance, and it benefits most from semantic filtering. Table III shows that accuracy increases from 93.6% (original) to 95.7% (light) and 97.4% (moderate). Macro-mAP rises from 0.969 to 0.984 and 0.991. Because only the training split is filtered, these gains reflect genuine improvements in generalisation rather than easier evaluation data. Embedding visualisations confirm that low-Q crops are also embedding outliers; removing them produces cleaner, linearly separable class clusters.

V. DISCUSSION

Across all pipelines, the results support a data-centric interpretation of AgroPest-12: the dataset's primary limitations stem from semantic noise, label inconsistency, and low-information crops rather than insufficient model capacity. Even strong architectures such as YOLOv8 and DINOv2 show clear improvements once the lowest-quality training samples are removed.

For the classical HOG+SVM pipeline, the gains from Q-filtering are small in absolute terms, yet consistent. This is expected: the features are not expressive enough to leverage improved data quality fully, but removing heavily blurred or background-dominated crops reduces variance and stabilises the classifier.

YOLOv8 exhibits a more interesting pattern. The lightly filtered dataset gives the best performance, while the moderate filter yields mixed results. This suggests a trade-off between removing harmful outliers (beneficial) and reducing species and appearance diversity (harmful). YOLOv8 in particular appears to benefit from difficult yet informative examples, which are partially removed in the moderate regime. The fact that moderate filtering still improves over the raw dataset confirms that a significant fraction of the original training set is genuinely detrimental.

The hybrid ViT/DINOv2 + linear classifier experiences the largest gains from Q-filtering. Because transformer embeddings cluster semantically similar insects tightly, low-quality crops manifest as clear embedding outliers. Removing them

Fig. 4. Macro-F1 for representative models under raw and Q-filtered regimes. Replace values with your measurements.

produces well-separated clusters and near-linear separability, which explains the substantial jump in accuracy and macro-mAP. This pipeline also benefits from reduced training time: with fewer samples but stronger semantic consistency, the classifier converges more quickly and reliably.

A consistent theme across all results is that semantic data quality is more influential than model architecture. Moderate changes to backbones, augmentations, or optimisation had much smaller effects than removing semantically inconsistent crops. This matches recent findings in data-centric ML and shows that, for challenging real-world agricultural datasets, careful curation can outperform architectural scaling.

A. Limitations

It is important to note that models trained on the filtered dataset are not directly comparable to models trained on the original raw dataset. Also, the proposed pipeline still suffers from a chicken-and-egg dependency on existing annotations. All quality estimates ultimately rely on the provided bounding boxes and class labels: if these are grossly incorrect or missing, Q can down-weight the affected samples but cannot fully repair or relabel them without human intervention. As a result, there remains a subset of incorrect or ambiguous labels that our filtering cannot resolve.

We cannot fully disentangle the contributions of individual nuisance factors such as blur, occlusion, clutter, and label noise. In real AgroPest-12 images these effects often co-occur, and our $Q(x)$ combines morphology and semantic consistency into a single scalar. As a result, we cannot cleanly attribute, for example, "2% mAP gain" to blur alone. The reported improvements should therefore be read as aggregate effects of filtering out obviously problematic crops, rather than precise attributions to specific image defects.

Q-filtering can over-prune rare or difficult cases. Although we enforce a per-class minimum when constructing filtered splits, hard-edge examples and rare morphologies are still more likely to have low $Q(x)$ and be removed. This may reduce the model's ability to generalise to truly challenging field conditions or unusual species appearances. Future work could mitigate this by soft-reweighting low-Q samples, adding a human-in-the-loop review stage for borderline cases, or explicitly modelling context (e.g., illumination, motion blur, host plant) rather than simply discarding such examples.

Our experiments focus on static images from a single dataset. In practice, pest monitoring systems operate under changing environmental conditions and must handle motion, wind, night-time imaging, and evolving species distributions. In that broader setting, "less is more" only holds if the data curation process is designed to preserve ecologically important variability while reducing harmful ambiguity and mislabelling. Our Q-score pipeline is a small step in that direction, showing that simple semantic- and morphology-aware heuristics can make existing models behave more stably on a noisy dataset, but it does not replace the need for richer, more diverse data in the long term.

Semantic filtering also risks removing precisely the hard edge cases that matter in real deployments. Rare species, unusual life stages, extreme poses, and difficult environmental conditions are more likely to receive low Q scores, and may be underrepresented or excluded in heavily filtered regimes. This improves model stability, increasing the risk of overfitting to a curated "easy" subset and degrading generalisation to other datasets or future field conditions. Per-class thresholds and human-in-the-loop review could mitigate this, although that defies the purpose of an automatic filtering.

VI. Conclusion and Future Work

This study compared three representative pipelines for insect detection and classification on AgroPest-12 and evaluated the impact of segmentation-assisted preprocessing and a semantic Quality Score based on morphology and DINOv2 embeddings. Across all models, semantic-aware filtering improved performance, but the magnitude varied widely.

Classical HOG+SVM improved only modestly, reflecting the limitations of handcrafted features in complex field conditions. YOLOv8 benefited more substantially, with the lightly filtered dataset giving the best results and demonstrating that moderate curation reduces harmful outliers without sacrificing diversity. The strongest gains occurred in the hybrid ViT/DINOv2 pipeline, where accuracy rose from 93.6% to 97.4% and macro-mAP from 0.969 to 0.991 under moderate filtering.

Future work will extend this framework along several directions. First, richer quality metrics could integrate temporal information, multi-view geometry, and texture-based cues, moving beyond static morphology and single-image semantics. Second, domain adaptation and self-supervised pretraining tailored to agricultural imagery may reduce the reliance on generic natural-image encoders and improve robustness to lighting, sensor, and crop variations. Third, quality-aware filtering strategies that preserve rare species and hard edge cases—potentially through human-in-the-loop review or active learning—are needed to balance benchmark performance against ecological coverage. Finally, deploying hybrid ViT-based embeddings and lightweight classifier heads on edge devices, possibly extended to short video sequences, offers a promising route toward real-time, data-aware pest monitoring systems that continuously curate and improve their own training sets.

Overall, these findings support a data-centric perspective: improving the semantic reliability of the training set yields larger and more stable gains than increasing model capacity. Future work should explore soft reweighting rather than hard filtering, quality-aware augmentation, and human-in-the-loop review for borderline cases to preserve rare and ecologically important examples while mitigating noise.

A. References

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